



SMART AI EMAIL BUILDER

A RAG-Based Client Email Generation System

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The Problem



1

Multiple Client Types

Enterprise accounts, small businesses, and long-term clients each expect a different tone — one template does not fit all.

2

Inconsistent Tone from Junior Staff

Junior team members often don't know how to phrase a sensitive reply, especially under pressure.

3

High-Stakes Moments

An angry, escalating client needs calm, precise wording fast — and mistakes here are costly.

*"One wrong email
can cost a
client relationship."*

The Solution



1

RAG-Grounded

Generation is grounded in a real company dataset of tone and scenarios — not generic AI output.

2

Gemini-Powered

Google's Gemini models handle both semantic retrieval and email generation.

3

Strictly Scoped

Two-layer guardrail keeps the tool to client emails only — everything else is refused.

4

Fully Auditable

Every prompt and every generated email is logged for admin review.

EXAMPLE

Input prompt

"The client is very upset because the delivery was delayed by one week. We need to calm the client down and request an additional three days to complete the remaining work."

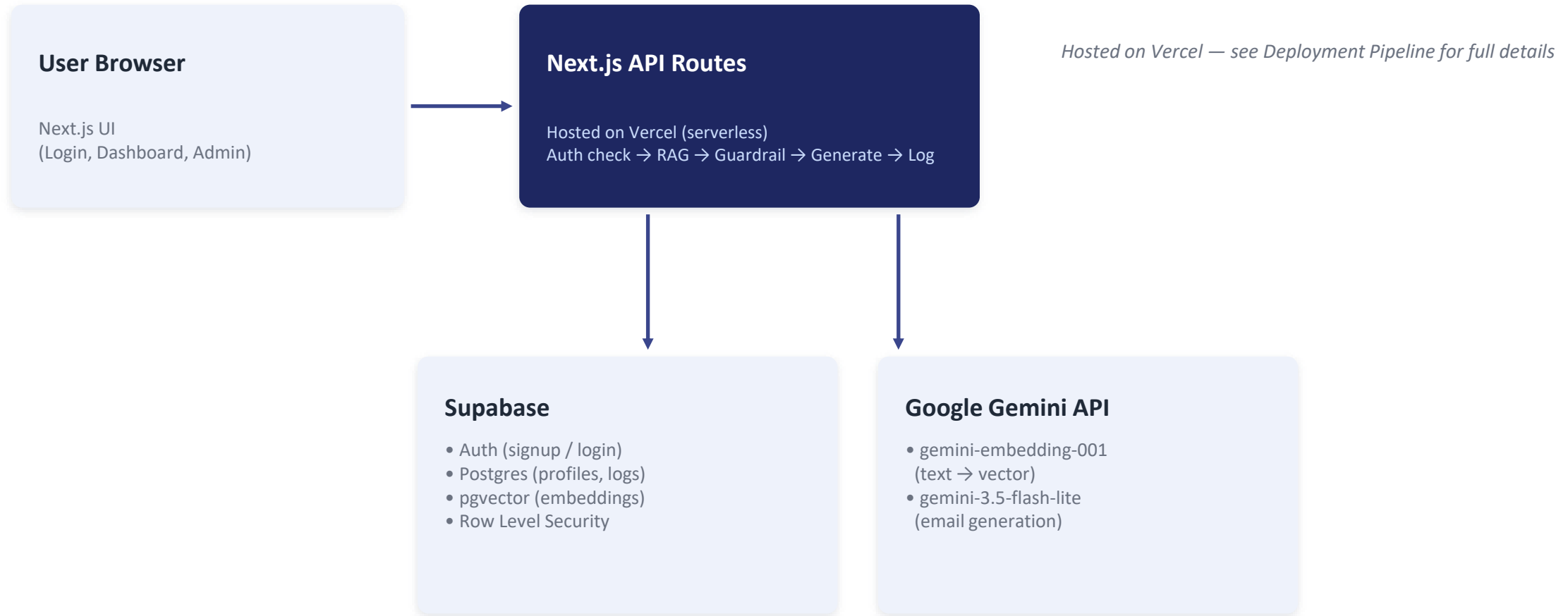


Generated email

Subject: Update on Project Timeline

I want to acknowledge upfront that we did not meet the timeline we committed to...

System Architecture



WHAT IT'S BUILT WITH

Technology Stack



N

Next.js 14

App Router, React Server Components, API routes

TS

TypeScript

Type-safe across frontend and backend

SB

Supabase

Postgres database, Auth, and pgvector extension

G

Gemini API

Embeddings + email generation (Google AI)

V

Vercel

Hosting, CI/CD, serverless functions

CSV

Dataset

Company tone/template source for RAG

Database Schema (Supabase / Postgres)

profiles

```
id (uuid, PK → auth.users)
email
username
is_admin (bool)
is_approved (bool)
created_at
```

→
1:many

prompt_logs

```
id (uuid, PK)
user_id (→ profiles.id)
username
input_prompt
output_email
matched_scenario
out_of_context (bool)
created_at
```

email_templates

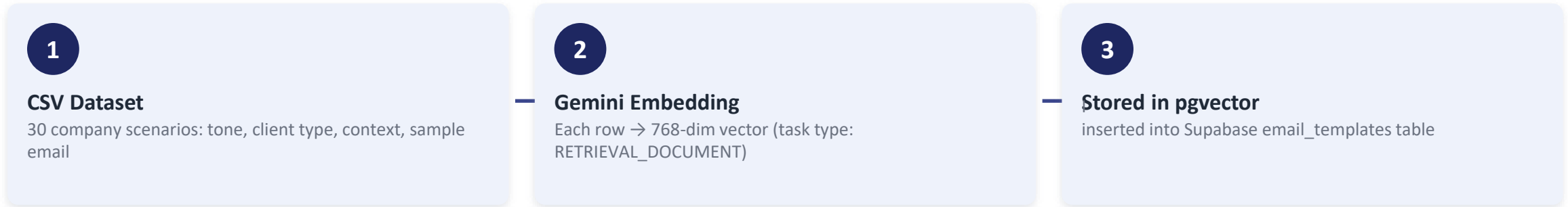
```
id (uuid, PK)
scenario
client_type
tone
context
sample_email
embedding (vector 768)
```

email_templates is used only by the RAG retrieval function — it has no foreign key relation to users.

Security: Row Level Security (RLS) is enabled on every table. Users can only read their own profile; only rows where `is_admin()` is true can read or update other users' data or view `prompt_logs`.

RAG Pipeline: Indexing + Retrieval

PHASE 1 — INDEXING (run once, or whenever the dataset is updated)



PHASE 2 — RUNTIME (every time a user submits a prompt)



Two-Layer Guardrail



1 Similarity Threshold

If the best-matching template's similarity score falls below the threshold, the request is rejected immediately — the Gemini generation call never even happens.

2 Strict System Prompt

Even when retrieval finds a weak match, Gemini itself is instructed: only write client emails, and refuse anything else with one fixed response.

Example — out-of-context prompt

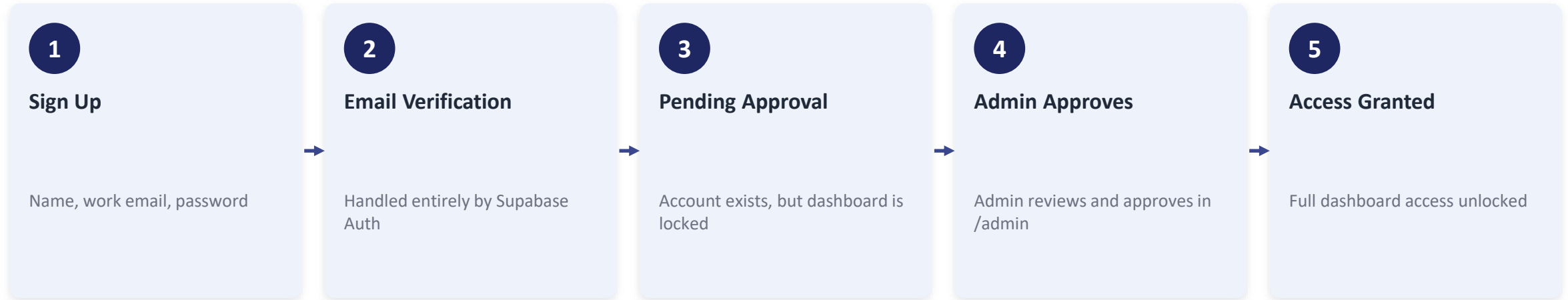
"Write me a poem about the moon"



Fixed refusal response

" I'm sorry, but this request is outside my scope. I can only help with drafting professional client emails."

Authentication & Approval Workflow



Admins can also revoke access at any time — flipping `is_approved` back to `false` immediately locks the user out of /dashboard on their next request, enforced by both middleware and the API route itself.

Admin Dashboard



Pending Signups

See every new signup waiting for approval, with one-click Approve



Approved Users

Full team list, with the ability to Revoke access instantly



Full Prompt Log

Every request: who sent it, the input prompt, the generated output, and which scenario matched

PROMPT LOG (LIVE VIEW)

Ali Raza

generated

Prompt: Angry client delayed delivery

Output: Update on Project Timeline...

Sana K.

generated

Prompt: Discount request from client

Output: Re: Pricing on Service...

Bilal A.

refused

Prompt: Write me a birthday poem

Output: Refused — out of context

Demo: The Angry Client Scenario



JUNIOR STAFF TYPES

"The client is very upset because the delivery was delayed by one week. We need to calm the client down and request an additional three days to complete the remaining work."

Matched scenario: "Angry client — project delayed"
Tone: calm, apologetic, confident, non-defensive

EMAIL BUILDER RESPONDS

Subject: Update on Project Timeline

Dear [Client Name],

I want to acknowledge upfront that we did not meet the timeline we committed to, and I understand the impact this has on your plans. That is on us...

...we are committing to [new specific date] as the delivery date.

Deployment Pipeline



COST

Vercel (Hobby tier) · Supabase (Free tier: Postgres + pgvector + Auth) · Gemini API (Free tier)

Current setup runs at zero hosting cost — suitable for internal team use at present scale.

Security Highlights



Row Level Security

Every table enforces RLS — users can only ever see their own profile; only admins can read `prompt_logs` or other users' profiles.

Service Role Isolation

The Supabase service role key is used only in server-side code; it is never sent to the browser.

Admin-Only API Routes

Every `/api/admin/*` route independently verifies `is_admin()` server-side before running — not just at the UI layer.

Managed Authentication

Passwords are handled entirely by Supabase Auth; the application database never stores or sees a plaintext password.

Roadmap



1

Email Notifications

Alert admins automatically when a new signup needs approval

2

Expanded Dataset

Add more real company scenarios as they come up in day-to-day work

3

Usage Analytics

Track most common scenarios, refusal rate, and team usage trends

4

Multi-language Tone

Support additional languages/dialects beyond formal English



Thank You

Questions & Discussion